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FINAL REPORT

TASK:

We compared the performance of three investment strategies (the two assigned, our custom 'Livermore Breakout Strategy' and in the visualizations, a fourth, 'Buy & Hold') over the historical daily stock data from the Alpaca Market Data API. The analysis uses data from June 1, 2021 to June 1, 2026 and evaluates each strategy using common portfolio performance metrics. Each strategy's portfolio is initially valued at $100,000 and implemented as long-only portfolios with no leverage or short selling.

STRATEGIES:

1. Trend Following Strategy

   The goal of this strategy is to capture sustained upward price movements using the Moving Average Convergence Divergence (MACD) indicator with the Average Directional Index (ADX). MACD identifies bullish momentum while ADX confirms that the market is trending strongly.

   Entry Rule: Buy when both MACD greater than MACD Signal Line AND ADX greater than 25.
   Exit Rule: Sell when MACD is less than the Signal Line as this indicates weakening momentum and possible trend reversal.

2. Mean Reversion Strategy

   This strategy assumes that prices temporarily deviate from their long-term average and eventually move back toward it. The strategy combines the Relative Strength Index (RSI) with Bollinger Bands to identify oversold buying opportunities.

   Entry Rule: Buy when RSI is less than 40 and closing price is below the lower Bollinger Band. (The asset may be temporarily oversold).
   Exit Rule: Sell when RSI is greater than 60 and the closing price is above the upper Bollinger Band, OR exit to flat when the price crosses back through the Bollinger Band middle line (20-day moving average), indicating that mean reversion has occurred.

3. Our Custom Strategy: "Livermore Breakout"

   We based our strategy on the Livermore Breakout strategy, entering positions only when all indicators confirm bullish conditions and using these three indicators:
   i. 20-day Simple Moving Average (SMA): <u>Tracks trend</u>

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ii. Williams %R: Tracks momentum
iii. Chaikin Money Flow (CMF): Tracks volume

Entry Rule: Buy when Closing price is greater than 20-day SMA short AND Williams %R greater than -50 AND CMF greater than 0.
Exit Rule: Exit to flat when any two out of the three conditions becomes bearish.
This goal is to avoid exiting on isolated indicator noise while reacting when multiple signals suggest weakening momentum.

4. Buy & Hold Strategy (Strategy used in the visualization comparisons)

This strategy purchases the selected stock at the beginning of the testing period and remains fully invested throughout the entire investment horizon. No trading decisions are made after the initial purchase.

Entry Rule: Buy on the first trading day.
Exit Rule: Hold the position until the end of the backtesting period.

PERFORMANCE COMPARISON

Ticker 1: AAPL

METRIC	**TOTAL RETURN(%)**	**SHARPE**	**SORTINO**	**MAX DRAWDOWN (%)**
STRATEGY				
Buy & Hold	151.09	0.81	1.18	-33.43
Trend Following	91.13	0.93	1.00	-17.75
Mean Reversion	55.45	0.64	0.46	-17.47
Custom Strategy	34.12	0.43	0.43	-30.12

Ticker 2: QQQ

METRIC	**TOTAL RETURN (%)**	**SHARPE**	**SORTINO**	**MAX DRAWDOWN**

				(%)
STRATEGY				
Buy & Hold	121.83	0.83	1.17	-35.62
Trend Following	38.01	0.59	0.56	-26.21
Mean Reversion	19.73	0.33	0.22	-16.09
Custom Strategy	27.56	0.43	0.48	-25.38

Ticker 3: MSFT

METRIC	**TOTAL RETURN (%)**	**SHARPE**	**SORTINO**	**MAX DRAWDOWN (%)**
STRATEGY				
Buy & Hold	81.99	0.59	0.84	-37.56
Trend Following	19.11	0.31	0.28	-33.96
Mean Reversion	69.53	0.82	0.64	-13.13
Custom Strategy	-26.09	-0.27	-0.24	-46.03

RESULTS

From the tablets, there is no single strategy that consistently outperformed the others across every stock. Performance varied depending on the characteristics of each asset, which shows that different trading strategies are better suited to different market conditions.

The Buy & Hold strategy serves as the base case of sorts by remaining fully invested throughout the testing period. While it generally captures the long-term appreciation of the selected asset, it also experiences the full impact of market downturns. It also consistently has the highest total return and sortino. Therefore this strategy, in these two respects, is the most profitable relative to risk. This demonstrates that remaining fully invested was the best approach during the strong long-term growth of the Nasdaq ETF. For the other two criteria, its Sharpe is within the top two amongst the tabulated tickers and Max Drawdown within the least two, exposing investors to greater portfolio volatility due to wide drawdown margins.

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The Trend Following strategy is to participate only during strong upward trends. By combining MACD with ADX, the strategy filters out many weak price movements and reduces exposure during periods without a clear trend by avoiding prolonged market declines. It does relatively well on the total returns for the first two tickers but not the third. For QQQ, for example, it achieved the highest Sharpe Ratio (0.93) and substantially reduced the maximum drawdown (-17.75% versus -33.43%), indicating superior risk-adjusted performance despite earning a lower return.

The Mean Reversion strategy is to profit from temporary price declines by purchasing assets that appear oversold. Because it waits for prices to recover toward their historical average, the strategy may perform well during sideways or range-bound markets but can struggle during prolonged downward trends when prices continue falling. For NVDA, for example, Mean Reversion experienced the smallest maximum drawdown (-16.09%), which shows its ability to reduce downside risk. While for MSFT, it delivered a comparable return (69.53%) while producing the highest Sharpe Ratio (0.82) and the lowest maximum drawdown (-13.13%). This suggests that Microsoft's relatively stable price movements created favorable opportunities for buying temporary pullbacks.

Our Custom Strategy combines momentum, trend confirmation, and volume-based money flow. Requiring agreement among multiple indicators reduces false signals and attempts to capture higher-quality trading opportunities. Its exit rule also requires multiple bearish conditions before closing a position, helping avoid unnecessary trades caused by short-term market noise. Across the tables, it produces mixed results. This shows that combining multiple indicators doesn’t necessarily improve performance unless the indicator combination aligns well with the market behavior of the underlying asset. For example, for MSFT, it generated a negative return (-26.09%), indicating that the combination of SMA, Williams %R, and Chaikin Money Flow was too restrictive or produced false signals during the price movements.

Ultimately, there is no universally optimal strategy. For example, a strategy producing slightly lower returns with substantially lower risk may be preferable for long-term investors; momentum-based strategies tend to perform best during strong, persistent trends, while mean-reversion approaches are more effective in stable or range-bound markets.